



Numerical Optimization

Winter semester 2025/26

Exercise sheet 3 for Numerical Optimization

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11. November 2025 - 89081 Ulm

Chapter 1

Solutions Exercise sheet 3

1.1 Exercise 1

1.1.1 Part a)

The statement asks us to prove Theorem 3.1.6, namely that if $\hat{x}$ is a local minimum of $f \in C^2(\Omega)$ with Ω open, then:

$$\nabla f(\hat{x}) = 0 \quad \text{and} \quad H_f(\hat{x}) \succeq 0,$$

where $\succeq 0$ means that the Hessian is positive semidefinite.

We proceed in two steps.

(1) $\nabla f(\hat{x}) = 0$

This is the first-order necessary condition. Intuitively, if the gradient were not zero, there would exist a direction of descent (opposite to the gradient) in which f decreases, contradicting the fact that $\hat{x}$ is a local minimum.

Formally, suppose that

$$\nabla f(\hat{x}) \neq 0.$$

Let

$$d = -\nabla f(\hat{x})$$

be the direction of steepest descent, and define the scalar function

$$\phi(t) = f(\hat{x} + td). \tag{1.1}$$

Since Ω is open, $\hat{x} + td \in \Omega$ for small t .

Using the multivariable chain rule:

$$\phi'(t) = \nabla f(\hat{x} + td) \cdot d. \tag{1.2}$$

In particular, at $t = 0$,

$$\phi'(0) = \nabla f(\hat{x}) \cdot d = \nabla f(\hat{x}) \cdot (-\nabla f(\hat{x})) = -\|\nabla f(\hat{x})\|^2 < 0. \tag{1.3}$$

Hence, for small $t > 0$, we have $\phi(t) < \phi(0)$, which implies

$$f(\hat{x} + td) < f(\hat{x}),$$

contradicting the local minimality of $\hat{x}$.

Therefore, the assumption is false, and we conclude that

$$\nabla f(\hat{x}) = 0.$$

(2) $H_f(\hat{x})$ is positive semidefinite

Now we use a **second-order argument**. Since the gradient vanishes at $\hat{x}$, the local behavior of f near $\hat{x}$ depends on its curvature, represented by the Hessian.

Let $v \in \mathbb{R}^n$ be an arbitrary direction, and define again a scalar function:

$$\phi(t) = f(\hat{x} + tv). \quad (1.4)$$

Geometrically, this means we are restricting f to the line passing through $\hat{x}$ in direction v .

Because $\hat{x}$ is a local minimum of f , it follows that $t = 0$ is a local minimum of ϕ . From basic single-variable calculus, for a C^2 function, if $t = 0$ is a local minimum, then

$$\phi'(0) = 0 \quad \text{and} \quad \phi''(0) \geq 0.$$

Compute the derivatives using the multivariable chain rule:

First derivative:

$$\phi'(t) = \nabla f(\hat{x} + tv) \cdot v. \quad (1.5)$$

Thus,

$$\phi'(0) = \nabla f(\hat{x}) \cdot v = 0, \quad (1.6)$$

since $\nabla f(\hat{x}) = 0$.

Second derivative: From multivariable calculus,

$$\phi''(0) = v^\top H_f(\hat{x}) v. \quad (1.7)$$

Because $t = 0$ is a local minimum of ϕ , we must have $\phi''(0) \geq 0$, hence

$$v^\top H_f(\hat{x}) v \geq 0 \quad \text{for all } v \in \mathbb{R}^n. \quad (1.8)$$

This is precisely the definition of a **positive semidefinite** matrix. Therefore,

$$H_f(\hat{x}) \succeq 0.$$

Geometric meaning of the multivariable chain rule as a dot product

The chain rule for $\phi(t) = f(\hat{x} + tv)$ can be written as

$$\frac{d}{dt} f(\hat{x} + tv) = \nabla f(\hat{x} + tv)^\top \frac{d(\hat{x} + tv)}{dt} = \nabla f(\hat{x} + tv)^\top v. \quad (1.9)$$

This shows that the derivative of f along a line is the *directional derivative* of f in direction v that is, the projection of the gradient onto v . In geometric terms, the gradient $\nabla f(\hat{x})$ points in the direction of steepest ascent, and the dot product $\nabla f(\hat{x}) \cdot v$ measures how much f increases when moving in direction v .

When $\nabla f(\hat{x}) = 0$, all directional derivatives vanish, which means f neither increases nor decreases in any direction exactly the condition for a stationary point.

Summary

Let $f \in C^2(\Omega)$ with Ω open, and let $\hat{x}$ be a local minimum. Then:

1. **(First order)** Since $\hat{x}$ is a local minimum and Ω is open,

$$\nabla f(\hat{x}) = 0.$$

2. **(Second order)** For any $v \in \mathbb{R}^n$, define $\phi(t) = f(\hat{x} + tv)$. Then $\phi'(0) = 0$ and $\phi''(0) = v^\top H_f(\hat{x}) v \geq 0$.

Hence, $H_f(\hat{x})$ is positive semidefinite. This completes the proof of Theorem 3.1.6.

1.1.2 Part b)

Goal: Find a C^2 function f and a point $\hat{x}$ such that

$$\nabla f(\hat{x}) = 0, \quad f \text{ has a strict local minimum at } \hat{x}, \quad H_f(\hat{x}) \text{ is not positive definite.}$$

Choice of the function

Consider

$$f(x, y) = x^2 + y^4, \quad (x, y) \in \mathbb{R}^2.$$

We will show that $\hat{x} = (0, 0)$ satisfies all requirements.

Gradient

The gradient is

$$\nabla f(x, y) = (2x, 4y^3).$$

At $(0, 0)$ we get

$$\nabla f(0, 0) = (0, 0),$$

so $(0, 0)$ is a critical point.

Hessian

The Hessian of f is

$$H_f(x, y) = \begin{pmatrix} 2 & 0 \\ 0 & 12y^2 \end{pmatrix}.$$

Evaluating at $(0, 0)$,

$$H_f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Test $v^\top H v$ explicitly

Let $v = (v_1, v_2)^\top \in \mathbb{R}^2$ be arbitrary. Then

$$v^\top H_f(0, 0)v = (v_1 \ v_2) \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = (v_1 \ v_2) \begin{pmatrix} 2v_1 \\ 0 \end{pmatrix} = 2v_1^2 \geq 0.$$

So the Hessian is *positive semidefinite*. It is *not* positive definite, because for example with $v = (0, 1)^\top$ we get

$$v^\top H_f(0, 0)v = 0.$$

To make this very concrete:

- For $v = (1, 0)^\top$: $v^\top H v = 2 > 0$.
- For $v = (0, 1)^\top$: $v^\top H v = 0$.

Hence $H_f(0, 0)$ fails to be positive *definite*, but it is positive *semidefinite*.

Show that $(0, 0)$ is a strict local minimum

For all (x, y) we have

$$f(x, y) = x^2 + y^4 \geq 0,$$

and

$$f(x, y) = 0 \iff x = 0 \text{ and } y = 0.$$

So $f(x, y) > 0$ for every $(x, y) \neq (0, 0)$, and $f(0, 0) = 0$. This means $(0, 0)$ is a *strict* local minimum.

We have found a C^2 function $f(x, y) = x^2 + y^4$ such that

$$\nabla f(0, 0) = 0, \quad (0, 0) \text{ is a strict local minimum,} \quad H_f(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \text{ is not positive definite.}$$

This shows that the converse of Theorem 3.1.5 is not true: having a strict local minimum does *not* force the Hessian to be positive definite.

1.1.3 Part c)

Goal: Find a C^2 function f and a point $\hat{x}$ such that

$$\nabla f(\hat{x}) = 0, \quad H_f(\hat{x}) \text{ is positive semidefinite,} \quad \text{but } f \text{ does not have a local minimum at } \hat{x}.$$

Choice of the function

Consider

$$f(x, y) = y^2 - x^4, \quad (x, y) \in \mathbb{R}^2.$$

We will again look at the point $\hat{x} = (0, 0)$.

Gradient

$$\nabla f(x, y) = (-4x^3, 2y),$$

so

$$\nabla f(0, 0) = (0, 0).$$

Hessian

The Hessian is

$$H_f(x, y) = \begin{pmatrix} -12x^2 & 0 \\ 0 & 2 \end{pmatrix}.$$

At $(0, 0)$,

$$H_f(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}.$$

Test $v^\top H v$

Let $v = (v_1, v_2)^\top$. Then

$$v^\top H_f(0, 0)v = (v_1 \ v_2) \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = (v_1 \ v_2) \begin{pmatrix} 0 \\ 2v_2 \end{pmatrix} = 2v_2^2 \geq 0.$$

Therefore $H_f(0, 0)$ is *positive semidefinite*.

Again, check with concrete directions:

- For $v = (0, 1)^\top$: $v^\top H v = 2 > 0$ (curvature upwards in the y -direction).
- For $v = (1, 0)^\top$: $v^\top H v = 0$ (flat in the x -direction at second order).

So the Hessian passes the “PSD” test, but is not positive definite.

Show that $(0, 0)$ is *not* a local minimum

Look at f restricted to the x -axis, i.e. set $y = 0$:

$$f(x, 0) = 0^2 - x^4 = -x^4.$$

For $x \neq 0$ we have $-x^4 < 0 = f(0, 0)$, so values of f near $(0, 0)$ can be made *strictly smaller* than $f(0, 0)$ just by moving in the x -direction. Hence $(0, 0)$ cannot be a local minimum.

This shows exactly the phenomenon the exercise wants: the Hessian is positive semidefinite, but this does *not* guarantee a local minimum, because along a direction in which the quadratic term vanishes (here the x -direction) the higher-order term (here $-x^4$) dominates and makes the function decrease.

The function

$$f(x, y) = y^2 - x^4$$

is C^2 , satisfies $\nabla f(0, 0) = 0$, and has a positive semidefinite Hessian at $(0, 0)$,

$$H_f(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix},$$

but $(0, 0)$ is *not* a local minimum, since $f(x, 0) = -x^4 < 0$ for $x \neq 0$.

Type of Hessian	What happens	Conclusion / Action
All eigenvalues > 0	Rises in all directions	Strict local minimum
All eigenvalues < 0	Falls in all directions	Strict local maximum
Positive and negative eigenvalues	Up in some, down in others	Saddle point (indefinite)
Some $= 0$, rest ≥ 0	Flat in some directions	Check higher-order terms
Some $= 0$, rest ≤ 0	Flat in some (downwards)	Check higher-order terms

Type	Condition	Interpretation
Positive definite (PD)	$v^\top H v > 0$ for all $v \neq 0$	Rises in all directions $\Rightarrow$ strict local minimum
Positive semidefinite (PSD)	$v^\top H v \geq 0$ for all v	Does not go down; may be flat in some directions
Negative definite (ND)	$v^\top H v < 0$ for all $v \neq 0$	Goes down in all directions $\Rightarrow$ strict local maximum
Negative semidefinite (NSD)	$v^\top H v \leq 0$ for all v	Does not go up; may be flat
Indefinite	There exist v_1, v_2 such that $v_1^\top H v_1 > 0$ and $v_2^\top H v_2 < 0$	Some directions up, some down $\Rightarrow$ saddle point

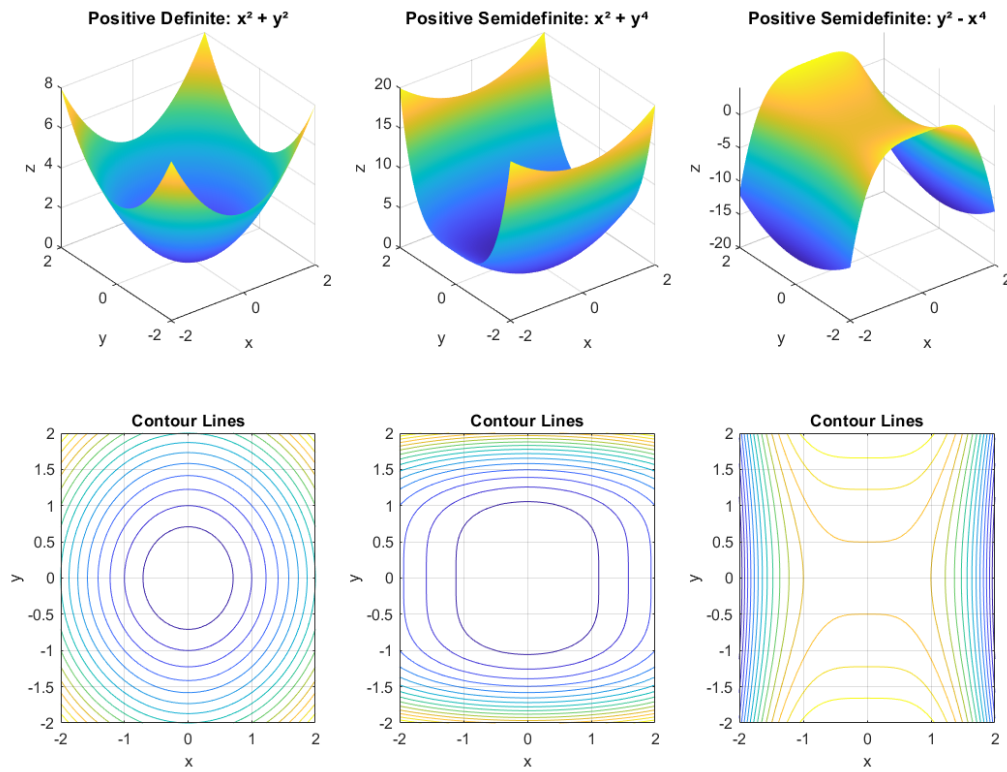


Figure 1.1: Description examples 1

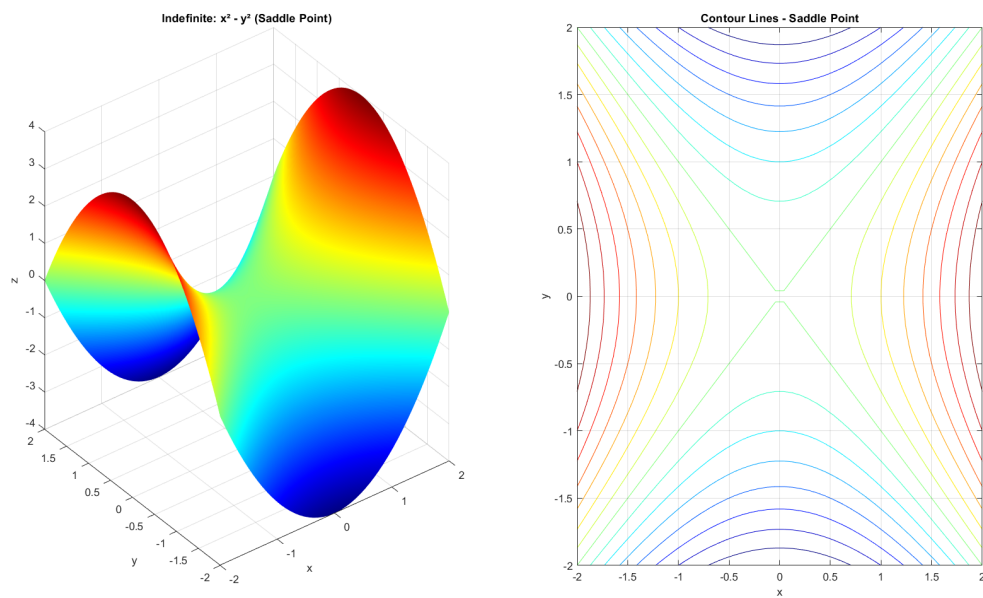


Figure 1.2: Description examples 2

1.2 Exercise 2

1.2.1 Part a)

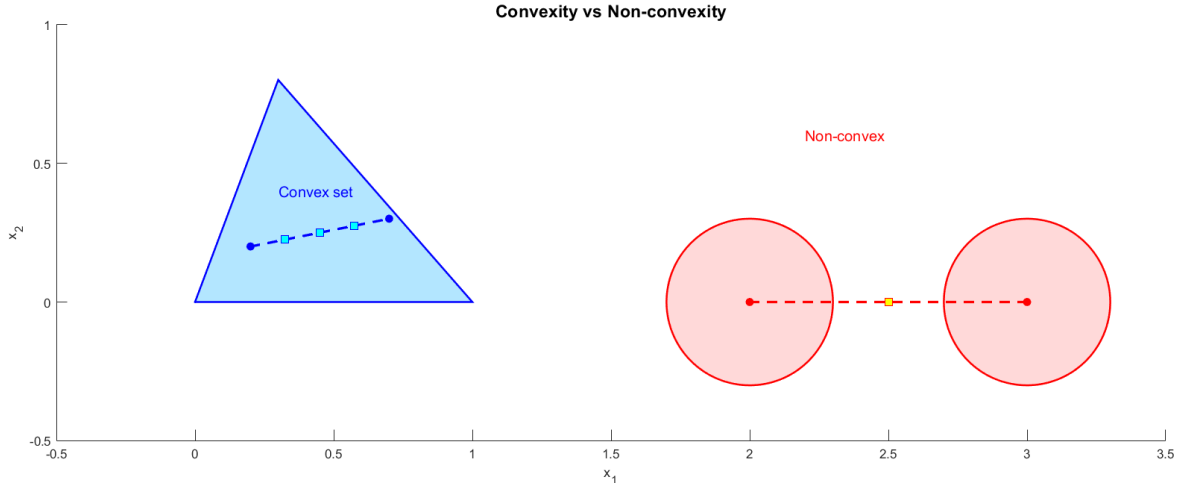


Figure 1.3: Explanation Convexity

Left (blue triangle) → Convex set.

Any two points chosen inside the triangle can be joined by a straight line, and the entire line segment remains inside the blue region. Mathematically, this means that every weighted average of those two points,

$$\alpha x + (1 - \alpha)y, \quad \text{with } \alpha \in [0, 1], \quad (1.10)$$

also belongs to the set.

Right (two red circles) → Non-convex set.

If one point is taken in the left circle and another in the right circle, the line connecting them passes through the empty space between the circles, which lies outside the set. Therefore, some convex combinations,

$$\alpha x + (1 - \alpha)y, \quad (1.11)$$

do not belong to the set, which means the set is *not convex*.

1.2.2 Part b)

Convex sets (2+6 points). A set $K \subset \mathbb{R}^n$ is called *convex* if

$$x, y \in K, \alpha \in [0, 1] \implies \alpha x + (1 - \alpha)y \in K. \quad (1.12)$$

We want to prove that a set $K \subset \mathbb{R}^n$ is convex *if and only if* it contains all convex combinations of its elements.

Recall that a point

$$x = \sum_{i=1}^m \lambda_i x_i \quad (1.13)$$

is a *convex combination* of $x_1, \dots, x_m$ if

$$\lambda_i \geq 0 \text{ for all } i, \quad \text{and} \quad \sum_{i=1}^m \lambda_i = 1. \quad (1.14)$$

(1) First direction

If K contains all convex combinations of its elements, then K is convex.

Proof. Assume that for any $x_1, \dots, x_m \in K$ and any coefficients $\lambda_i \geq 0$ with $\sum \lambda_i = 1$, the point $\sum \lambda_i x_i$ also belongs to K . Now take only two points $x, y \in K$ and any $\alpha \in [0, 1]$. Then

$$z = \alpha x + (1 - \alpha)y \quad (1.15)$$

is a convex combination of the two points x and y (here $m = 2$, $\lambda_1 = \alpha$, $\lambda_2 = 1 - \alpha$). By assumption, $z \in K$. This is exactly the definition of a convex set:

$$x, y \in K, \alpha \in [0, 1] \Rightarrow \alpha x + (1 - \alpha)y \in K. \quad (1.16)$$

Hence K is convex.

(2) direction:

If K is convex, then it contains all convex combinations of its elements.

We proceed by induction on the number of points m .

Step 0 (Induction goal). We show that for all $m \geq 2$: if $x_1, \dots, x_m \in K$ and $\lambda_1, \dots, \lambda_m \geq 0$ with $\sum_{i=1}^m \lambda_i = 1$, then

$$x := \sum_{i=1}^m \lambda_i x_i \in K. \quad (1.17)$$

Base case ($m = 2$). Let $x_1, x_2 \in K$ and $\lambda_1, \lambda_2 \geq 0$ with $\lambda_1 + \lambda_2 = 1$. By definition of convexity,

$$\lambda_1 x_1 + \lambda_2 x_2 \in K. \quad (1.18)$$

Thus, the base case holds.

Case $m = 3$. Let $x_1, x_2, x_3 \in K$ and $\lambda_1, \lambda_2, \lambda_3 \geq 0$ with $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

If $\lambda_3 = 0$, then

$$x = \lambda_1 x_1 + \lambda_2 x_2, \quad (1.19)$$

which is already in K by the base case.

If $\lambda_3 > 0$, define

$$s := \lambda_1 + \lambda_2 = 1 - \lambda_3 > 0, \quad (1.20)$$

and consider

$$y := \frac{\lambda_1}{s} x_1 + \frac{\lambda_2}{s} x_2. \quad (1.21)$$

The coefficients $\frac{\lambda_1}{s}, \frac{\lambda_2}{s}$ are nonnegative and sum to 1, hence y is a convex combination of x_1 and x_2 . By the case $m = 2$, we have $y \in K$.

Now rewrite x as

$$x = \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 = s \left(\frac{\lambda_1}{s} x_1 + \frac{\lambda_2}{s} x_2 \right) + \lambda_3 x_3 = s y + \lambda_3 x_3. \quad (1.22)$$

Since $y, x_3 \in K$, $s, \lambda_3 \geq 0$, and $s + \lambda_3 = 1$, the point x is a convex combination of two points of K . By convexity, $x \in K$. Therefore, the statement holds for $m = 3$.

Inductive step (general m). Assume that for some $m - 1 \geq 2$, any convex combination of $m - 1$ points of K lies in K . We prove it for m .

Let $x_1, \dots, x_m \in K$ and $\lambda_1, \dots, \lambda_m \geq 0$ with $\sum_{i=1}^m \lambda_i = 1$.

If $\lambda_m = 0$, then

$$x = \sum_{i=1}^m \lambda_i x_i = \sum_{i=1}^{m-1} \lambda_i x_i, \quad (1.23)$$

which is a convex combination of $m - 1$ points; hence $x \in K$ by the induction hypothesis.

If $\lambda_m > 0$, define

$$s := \sum_{i=1}^{m-1} \lambda_i = 1 - \lambda_m > 0, \quad y := \sum_{i=1}^{m-1} \frac{\lambda_i}{s} x_i. \quad (1.24)$$

Then $\frac{\lambda_i}{s} \geq 0$ and $\sum_{i=1}^{m-1} \frac{\lambda_i}{s} = 1$, so by the induction hypothesis $y \in K$.

Finally,

$$x = \sum_{i=1}^m \lambda_i x_i = s y + \lambda_m x_m. \quad (1.25)$$

Since $y, x_m \in K$ and $s, \lambda_m \geq 0$ with $s + \lambda_m = 1$, by convexity we conclude that $x \in K$.

Thus, the statement holds for all $m \geq 2$.

Example

Consider the full triangle with vertices

$$x_1 = (0, 0), \quad x_2 = (1, 0), \quad x_3 = (0, 1). \quad (1.26)$$

This set K is convex (it is a filled triangle).

Choose coefficients

$$\lambda_1 = 0.2, \quad \lambda_2 = 0.5, \quad \lambda_3 = 0.3, \quad (1.27)$$

which satisfy $\lambda_i \geq 0$ and $\lambda_1 + \lambda_2 + \lambda_3 = 1$.

(1) Compute the sum of the first two weights:

$$s = \lambda_1 + \lambda_2 = 1 - \lambda_3 = 0.7. \quad (1.28)$$

(2) Compute an intermediate point y using only x_1 and x_2 :

$$y = \frac{\lambda_1}{s} x_1 + \frac{\lambda_2}{s} x_2 = \frac{0.2}{0.7} (0, 0) + \frac{0.5}{0.7} (1, 0) = (0.714 \dots, 0). \quad (1.29)$$

This point y lies on the segment joining x_1 and x_2 . Since K is convex, $y \in K$.

(3) Combine y with x_3 :

$$x = s y + \lambda_3 x_3. \quad (1.30)$$

Substituting the values,

$$x = 0.7(0.714, 0) + 0.3(0, 1) = (0.5, 0.3). \quad (1.31)$$

The point $(0.5, 0.3)$ is inside the triangle K . This confirms numerically that a convex combination of the three points of K remains in K , illustrating the general result.

Both directions have been proved. A set K is convex if and only if it contains all convex combinations of its elements.

1.3 Exercise 3

1.3.1 Part a)

Why does K have to be convex in order to define a convex function $f : K \rightarrow \mathbb{R}$?

The convexity inequality involves points of the form

$$z = \lambda x + (1 - \lambda)y, \quad \lambda \in [0, 1]. \quad (1.32)$$

If $x, y \in K$, the definition of a convex function requires evaluating $f(z)$ at this z . However, if K is not convex, the point z may lie outside of K , and in that case $f(z)$ is not even defined.

Short idea: To compare “the value of the function at the endpoints” and “the value at the convex combination of the points,” those combinations must remain inside the domain. Therefore, the domain K itself must be convex.

Example:

Let

$$K = \{0, 2\} \subset \mathbb{R}, \quad (1.33)$$

which is not convex. Choose $x = 0$, $y = 2$, and $\lambda = 0.5$. Then

$$z = \lambda x + (1 - \lambda)y = 1, \quad (1.34)$$

but $1 \notin K$. Hence, we cannot even write the convexity inequality for this set.

1.3.2 Part b)

Example of a convex function $f : \mathbb{R} \rightarrow \mathbb{R}$

Consider the classical example:

$$f(x) = x^2. \quad (1.35)$$

Its second derivative is

$$f''(x) = 2 > 0, \quad (1.36)$$

which directly implies that f is convex on $\mathbb{R}$.

Numerical illustration:

Let $x = 0$, $y = 2$, and $\lambda = 0.3$.

$$f(0) = 0, \quad f(2) = 4. \quad (1.37)$$

Intermediate point:

$$z = \lambda x + (1 - \lambda)y = 0.3 \cdot 0 + 0.7 \cdot 2 = 1.4, \quad (1.38)$$

$$f(1.4) = 1.96. \quad (1.39)$$

Right-hand side of the inequality:

$$\lambda f(x) + (1 - \lambda)f(y) = 0.3 \cdot 0 + 0.7 \cdot 4 = 2.8. \quad (1.40)$$

We compare:

$$f(1.4) = 1.96 \leq 2.8, \quad (1.41)$$

so the convexity condition holds. Geometrically, the point on the curve lies below the line segment joining $(0, 0)$ and $(2, 4)$.

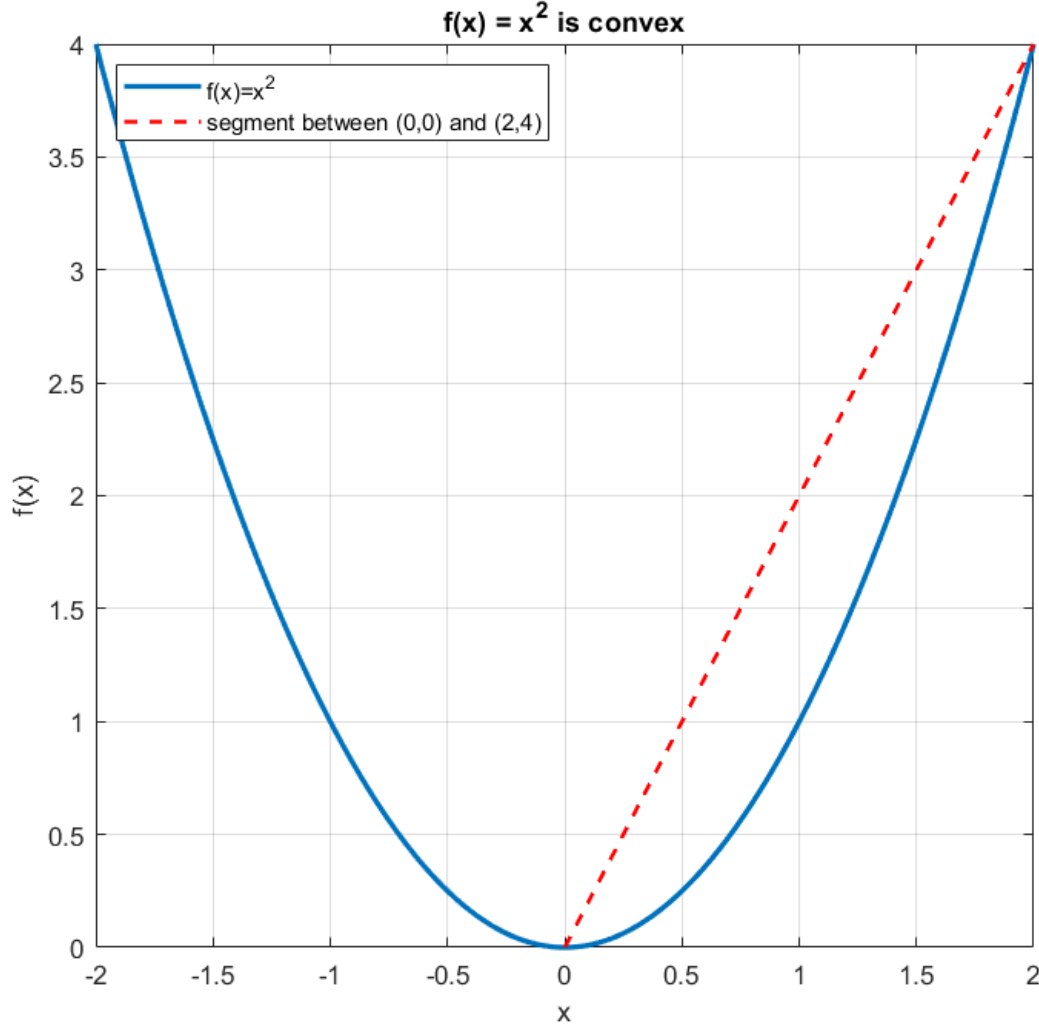


Figure 1.4: Convex Function

Definition Convexity of some basic functions A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called *convex* if for any $x, y \in \mathbb{R}^n$ and any $\lambda \in [0, 1]$ we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.42)$$

Intuitively, the value of the function at the point between x and y (i.e. at the convex combination $\lambda x + (1 - \lambda)y$) is not larger than the straight line joining $f(x)$ and $f(y)$.

1.3.3 Part c)

Affine function $f(x) = a^T x + b$ is **convex**.

Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Take $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$. Then

$$f(\lambda x + (1 - \lambda)y) = a^T(\lambda x + (1 - \lambda)y) + b. \quad (1.43)$$

Using linearity of the dot product,

$$a^T(\lambda x + (1 - \lambda)y) = \lambda a^T x + (1 - \lambda)a^T y, \quad (1.44)$$

so

$$f(\lambda x + (1 - \lambda)y) = \lambda a^T x + (1 - \lambda)a^T y + b. \quad (1.45)$$

On the other hand,

$$\lambda f(x) + (1 - \lambda)f(y) = \lambda(a^T x + b) + (1 - \lambda)(a^T y + b) = \lambda a^T x + (1 - \lambda)a^T y + b. \quad (1.46)$$

Thus the two sides are equal:

$$f(\lambda x + (1 - \lambda)y) = \lambda f(x) + (1 - \lambda)f(y), \quad (1.47)$$

which certainly implies the $\leq$ condition. Hence every affine function is convex.

1.3.4 Part d)

Quadratic function $f(x) = \frac{1}{2}x^T Qx + p^T x + q$ **is convex if** $Q \succeq 0$.

Let $Q \in \mathbb{R}^{n \times n}$ be symmetric, $p \in \mathbb{R}^n$, $q \in \mathbb{R}$, and define

$$f(x) = \frac{1}{2}x^T Qx + p^T x + q. \quad (1.48)$$

We recall a standard result: if f is twice differentiable and its Hessian $\nabla^2 f(x)$ is positive semidefinite for all x , then f is convex.

Compute the gradient. Since Q is symmetric,

$$\nabla f(x) = Qx + p. \quad (1.49)$$

Differentiate once more to get the Hessian:

$$\nabla^2 f(x) = Q. \quad (1.50)$$

This Hessian does not depend on x . If the assumption $Q \succeq 0$ holds (i.e. $z^T Qz \geq 0$ for all $z \in \mathbb{R}^n$), then the Hessian is positive semidefinite everywhere, so f is convex.

Note that the terms $p^T x$ (affine) and q (constant) do not destroy convexity; the convexity is determined by the quadratic term $\frac{1}{2}x^T Qx$ and the condition $Q \succeq 0$.

1.3.5 Part e)

The function $f(x) = \|x - a\|_2$ **is convex but not strictly convex.**

Let $a \in \mathbb{R}^n$ be fixed and define

$$f(x) = \|x - a\|_2. \quad (1.51)$$

Take $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$. Then

$$f(\lambda x + (1 - \lambda)y) = \|\lambda x + (1 - \lambda)y - a\| = \|\lambda(x - a) + (1 - \lambda)(y - a)\|. \quad (1.52)$$

Using the triangle inequality and homogeneity of the Euclidean norm,

$$\|\lambda(x - a) + (1 - \lambda)(y - a)\| \leq \|\lambda(x - a)\| + \|(1 - \lambda)(y - a)\| = \lambda\|x - a\| + (1 - \lambda)\|y - a\|. \quad (1.53)$$

Thus,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad (1.54)$$

so f is convex.

It is, however, *not* strictly convex. For example, take $u \neq 0$ and consider

$$x = a + u, \quad y = a + 2u, \quad \lambda = \frac{1}{2}. \quad (1.55)$$

Then

$$f(x) = \|u\|, \quad f(y) = 2\|u\|, \quad f\left(\frac{1}{2}x + \frac{1}{2}y\right) = f\left(a + \frac{3}{2}u\right) = \frac{3}{2}\|u\|, \quad (1.56)$$

and

$$\frac{1}{2}f(x) + \frac{1}{2}f(y) = \frac{1}{2}\|u\| + \frac{1}{2} \cdot 2\|u\| = \frac{3}{2}\|u\|. \quad (1.57)$$

So we have equality instead of a strict inequality, which shows that f fails to be strictly convex.

1.4 Exercise 4 : Compositions of Convex Functions

Let $C \subseteq \mathbb{R}^n$ be convex. We want to prove several statements about compositions of convex functions.

Definition (Convex function)

A function $f : C \rightarrow \mathbb{R}$ is said to be **convex** if for all $x, y \in C$ and every $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.58)$$

Geometrically, this means that the graph of f lies below the straight line segment joining the points $(x, f(x))$ and $(y, f(y))$.

1.4.1 Part a)

Linear combinations of convex functions

Let $f_i : C \rightarrow \mathbb{R}$ be convex for $i = 1, \dots, m$ and $\beta_i \geq 0$. Define

$$g(x) = \sum_{i=1}^m \beta_i f_i(x). \quad (1.59)$$

i) Show that g is convex on C **Intuitive idea:** If each f_i is convex and all coefficients β_i are nonnegative, then the convexity is preserved when taking their weighted sum. It behaves like averaging convex bowls the shape remains convex.

Proof step by step: Take arbitrary $x, y \in C$ and $\lambda \in [0, 1]$. Then:

$$g(\lambda x + (1 - \lambda)y) = \sum_{i=1}^m \beta_i f_i(\lambda x + (1 - \lambda)y). \quad (1.60)$$

Since each f_i is convex, we can write:

$$f_i(\lambda x + (1 - \lambda)y) \leq \lambda f_i(x) + (1 - \lambda)f_i(y). \quad (1.61)$$

Substitute this inequality into the expression for g :

$$g(\lambda x + (1 - \lambda)y) \leq \sum_{i=1}^m \beta_i [\lambda f_i(x) + (1 - \lambda)f_i(y)]. \quad (1.62)$$

Now distribute λ and $(1 - \lambda)$ over the sum:

$$g(\lambda x + (1 - \lambda)y) \leq \lambda \sum_{i=1}^m \beta_i f_i(x) + (1 - \lambda) \sum_{i=1}^m \beta_i f_i(y). \quad (1.63)$$

Recognizing the definition of $g(x)$ and $g(y)$, we get:

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y). \quad (1.64)$$

Thus, g is convex on C . □

(ii) Strict convexity Assume that for some fixed $k \in \{1, \dots, m\}$, f_k is strictly convex and $\beta_k > 0$.

Recall: A function f_k is *strictly convex* if

$$f_k(\lambda x + (1 - \lambda)y) < \lambda f_k(x) + (1 - \lambda)f_k(y) \quad \text{for all } x \neq y, \lambda \in (0, 1). \quad (1.65)$$

As before, we sum all inequalities for f_i . For all $i \neq k$, we have $\leq$, but for $i = k$, the inequality is strict $<$. Then:

$$g(\lambda x + (1 - \lambda)y) = \sum_{i=1}^m \beta_i f_i(\lambda x + (1 - \lambda)y) < \lambda g(x) + (1 - \lambda)g(y). \quad (1.66)$$

Hence, g is strictly convex on C . □

1.4.2 Part b)

Let $g : C \rightarrow \mathbb{R}^m$ be an affine mapping:

$$g(x) = Ax + b, \quad (1.67)$$

with $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

(i) Show that $g(C)$ is convex Take any $x_1, x_2 \in C$ and $\lambda \in [0, 1]$. Because C is convex, the point $\lambda x_1 + (1 - \lambda)x_2$ belongs to C . Apply g :

$$g(\lambda x_1 + (1 - \lambda)x_2) = A(\lambda x_1 + (1 - \lambda)x_2) + b \quad (1.68)$$

$$= \lambda Ax_1 + (1 - \lambda)Ax_2 + b \quad (1.69)$$

$$= \lambda(Ax_1 + b) + (1 - \lambda)(Ax_2 + b) \quad (1.70)$$

$$= \lambda g(x_1) + (1 - \lambda)g(x_2). \quad (1.71)$$

Hence, $\lambda g(x_1) + (1 - \lambda)g(x_2) \in g(C)$, which means that $g(C)$ is convex. □

(ii) Show that if $f : g(C) \rightarrow \mathbb{R}$ is convex, then $f \circ g$ is convex on C Let $h(x) = f(g(x))$. We want to verify the convexity condition for h .

Take $x, y \in C$ and $\lambda \in [0, 1]$. Then:

$$h(\lambda x + (1 - \lambda)y) = f(g(\lambda x + (1 - \lambda)y)). \quad (1.72)$$

Since g is affine,

$$g(\lambda x + (1 - \lambda)y) = \lambda g(x) + (1 - \lambda)g(y). \quad (1.73)$$

Applying the convexity of f :

$$f(\lambda g(x) + (1 - \lambda)g(y)) \leq \lambda f(g(x)) + (1 - \lambda)f(g(y)). \quad (1.74)$$

Combining both results:

$$h(\lambda x + (1 - \lambda)y) \leq \lambda h(x) + (1 - \lambda)h(y). \quad (1.75)$$

Therefore, $f \circ g$ is convex on C . □

1.4.3 Part c)

Maximum of a convex function and zero Let $f : C \rightarrow \mathbb{R}$ be convex, and define

$$g(x) = \max\{0, f(x)\}. \quad (1.76)$$

Intuitive idea: The pointwise maximum of convex functions is convex. Here we consider two convex functions:

$$f_1(x) = 0 \quad \text{and} \quad f_2(x) = f(x).$$

Formal proof: For any $x, y \in C$ and $\lambda \in [0, 1]$, we have:

$$g(\lambda x + (1 - \lambda)y) = \max\{0, f(\lambda x + (1 - \lambda)y)\} \quad (1.77)$$

$$\leq \max\{0, \lambda f(x) + (1 - \lambda)f(y)\} \quad (1.78)$$

$$\leq \lambda \max\{0, f(x)\} + (1 - \lambda) \max\{0, f(y)\}. \quad (1.79)$$

The last inequality follows from the convexity of the maximum operator. Hence, g is convex on C . $\square$

1.4.4 Part d)

Square of a nonnegative convex function Let $f : C \rightarrow \mathbb{R}$ be convex and nonnegative. Define

$$g(x) = [f(x)]^2. \quad (1.80)$$

Intuitive idea: The squaring function $\phi(t) = t^2$ is convex and nondecreasing for $t \geq 0$. There is a useful composition rule:

If f is convex and ϕ is convex and nondecreasing, then $\phi \circ f$ is convex.

Proof: Since $f(x) \geq 0$, we can apply this rule directly. For any $x, y \in C$ and $\lambda \in [0, 1]$, convexity of f gives:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.81)$$

Now apply $\phi(t) = t^2$ to both sides (which preserves the inequality since ϕ is nondecreasing for $t \geq 0$):

$$[f(\lambda x + (1 - \lambda)y)]^2 \leq [\lambda f(x) + (1 - \lambda)f(y)]^2. \quad (1.82)$$

By expanding the right-hand side:

$$[\lambda f(x) + (1 - \lambda)f(y)]^2 = \lambda^2 f(x)^2 + (1 - \lambda)^2 f(y)^2 + 2\lambda(1 - \lambda)f(x)f(y). \quad (1.83)$$

Although this expression looks more complicated, one can show it satisfies

$$[f(\lambda x + (1 - \lambda)y)]^2 \leq \lambda f(x)^2 + (1 - \lambda)f(y)^2,$$

because $f(x), f(y) \geq 0$ and $\phi(t) = t^2$ is convex. Hence,

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y), \quad (1.84)$$

so g is convex on C . $\square$

Summary

Part	Definition	Result	Main idea
(a)(i)	$g = \sum_i \beta_i f_i$	Convex	Weighted sum of convex functions
(a)(ii)	One f_k strictly convex, $\beta_k > 0$	Strictly convex	One strict term makes sum strict
(b)(i)	$g(x) = Ax + b$	$g(C)$ convex	Affine image of convex set is convex
(b)(ii)	$f \circ g$	Convex	Composition of convex with affine map
(c)	$g = \max\{0, f\}$	Convex	Maximum of convex functions is convex
(d)	$g = f^2, f \geq 0$	Convex	t^2 is convex and nondecreasing

Exercise sheet 3 for Numerical Optimization

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Deadline: November 5, 2025

Exercise 1 (Second-order necessary optimality condition, 7+2+2 points)

- a) Prove theorem 3.1.6 in the lecture notes. That is if $f \in C^2(\Omega, \mathbb{R})$ on an open set $\Omega \subset \mathbb{R}^n$ attains a local minimum at $\hat{x} \in \Omega$, then it holds that $\nabla f(\hat{x}) = \mathbf{0}$ and $H_f(\hat{x})$ is positive semidefinite.

Note: A matrix $A \in \mathbb{R}^{n \times n}$ is positive semidefinite if

$$v^T A v \geq 0 \quad \forall v \in \mathbb{R}^n .$$

It is positive definite if

$$v^T A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{\mathbf{0}\} .$$

If $f \in C^2(\Omega, \mathbb{R})$ satisfies $\nabla f(\hat{x}) = \mathbf{0}$ at $\hat{x} \in \Omega$ the following two statements hold:

- Theorem 3.1.5: If $H_f(\hat{x})$ is positive definite, then f has a strict local minimum at $\hat{x}$.
- Theorem 3.1.6: If f has a local minimum at $\hat{x}$, $H_f(\hat{x})$ is positive semidefinite.

We want to find examples in order to see that the statements' converses are not true.

- b) Find a C^2 -function f and a point $\hat{x}$ with $\nabla f(\hat{x}) = \mathbf{0}$ that has a strict local minimum at $\hat{x}$, but $H_f(\hat{x})$ is not positive definite.
- c) Find a C^2 -function f and a point $\hat{x}$ with $\nabla f(\hat{x}) = \mathbf{0}$ and positive semidefinite $H_f(\hat{x})$, but f does not have a local minimum at $\hat{x}$.

Exercise 2 (Convex sets, 2+6 points)

A set $K \subset \mathbb{R}^n$ is called *convex* if

$$x, y \in K, \alpha \in [0, 1] \implies \alpha x + (1 - \alpha)y \in K . \quad (1)$$

a) Draw a two-dimensional convex set and a two-dimensional non-convex set and explain.

b) Let $x_1, \dots, x_m \in \mathbb{R}^n$, then

$$x = \sum_{i=1}^m \lambda_i x_i$$

is a *convex combination* of $x_1, \dots, x_m$, if $\lambda_i \geq 0$ for $i = 1, \dots, m$ and $\sum_{i=1}^m \lambda_i = 1$.

Prove that a set is convex if and only if it contains all convex combinations of its elements.

Note: Prove *convex* $\Rightarrow$ *contains all convex combinations* using induction over m .

Exercise 3 (Convex functions, 2+2+3+4+4 points)

Let $K \subset \mathbb{R}^n$ be convex and $K \neq \emptyset$. A function $f : K \rightarrow \mathbb{R}$ is *convex* on K , if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in K$ and $\lambda \in [0, 1]$. It is *strict convex* if

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in K$ with $x \neq y$ and $\lambda \in (0, 1)$.

- a) Why does K have to be convex in order to define a convex function $f : K \rightarrow \mathbb{R}$?
- b) Find a function mapping from $\mathbb{R}$ to $\mathbb{R}$ that is convex and draw it. Explain, how convexity can be illustrated in this simple example.
- c) Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. Show that the affine function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = a^T x + b$$

is convex.

- d) Let $Q \in \mathbb{R}^{n \times n}$ be symmetric and $p \in \mathbb{R}^n, q \in \mathbb{R}$. Show that the quadratic function

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \frac{1}{2}x^T Q x + p^T x + q$$

is convex if Q is positive semidefinite.

- e) Let $a \in \mathbb{R}^n$. Prove that

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \|x - a\|_2$$

is convex, but not strictly convex.

Exercise 4 (Compositions of convex functions, 4+4+4+4 points)

Let $C \in \mathbb{R}^n$ be convex. Prove the following statements about compositions of convex functions.

a) Let $f_i : C \rightarrow \mathbb{R}$ be convex and $\beta_i \geq 0$ for $i = 1, \dots, m$.

i) Show that

$$g(x) := \sum_{i=1}^m \beta_i f_i(x)$$

is convex on C .

ii) Additionally, suppose that for an arbitrary but fixed $k \in \{1, \dots, m\}$, f_k is strictly convex and $\beta_k > 0$. Prove that then $g(x)$ is strictly convex on C .

b) For an affine function

$$g : C \rightarrow \mathbb{R}^m, \quad g(x) := Ax + b \quad \text{with} \quad A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m$$

show that

i) the set $g(C)$ is convex.

ii) Also show that if $f : g(C) \rightarrow \mathbb{R}$ is convex, $f \circ g$ is also convex on C .

c) For $f : C \rightarrow \mathbb{R}$ convex, show that

$$g(x) := \max\{0, f(x)\}$$

is also convex on C .

d) For $f : C \rightarrow \mathbb{R}$ convex and non-negative, show that

$$g(x) := f(x)^2$$

is also convex on C .